

# **Development and Implementation of a Loyalty System for a Confectionery Based on Smart Contracts**

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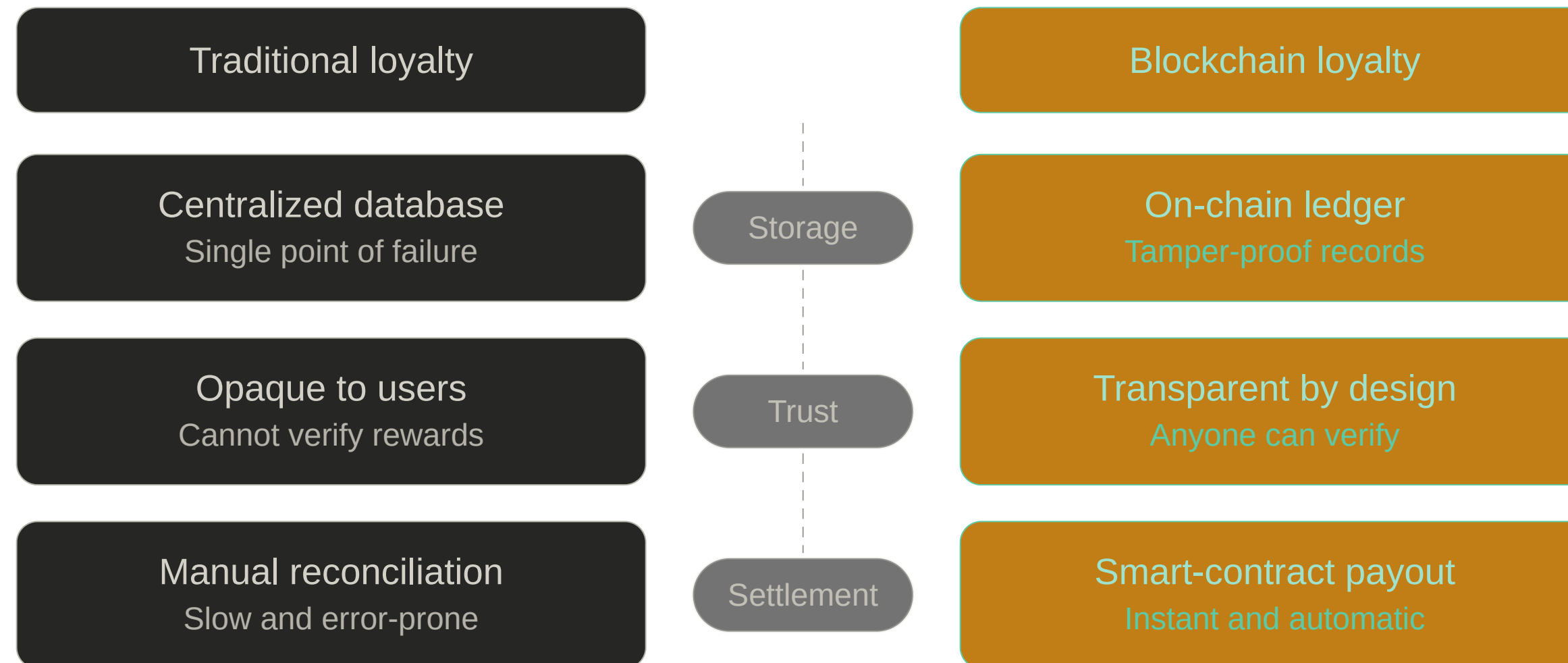
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# INTRODUCTION: PROBLEM STATEMENT



**Figure 1:** Comparison of traditional and Blockchain-based loyalty systems.

## Challenges of traditional loyalty systems:

- Centralized reward storage
- Limited transparency
- Risk of data manipulation
- Difficult partner integration
- High administrative effort

## How to make loyalty programs transparent, secure, and user-friendly?

Blockchain technology enables transparent, auditable, and tamper-resistant reward management.

# INTRODUCTION

## PURPOSE

Develop a blockchain-based loyalty platform for confectionery businesses using smart contracts and Telegram Mini Apps.

## OBJECTIVES

- Analyze existing loyalty solutions •
- Design system architecture •
- Develop smart contracts •
- Implement Telegram Mini App •
- Evaluate customer and business benefits •

## RELEVANCE

- “55% of loyalty members don't know their points balance, and 41% are unaware of their points value”. (Bond Brand Loyalty, 2017)
- “Acquiring a new customer costs 5–25 times more than retaining one, driving high administrative overhead”. (Plaisir, M. L., & Team, A., 2024)
- “Only 48.6% of earned loyalty points are ever redeemed, customers can't independently track or verify them” (Garai T., 2023)

## NOVELTY

- TON blockchain integration •
- Telegram Mini App interface •
- Gasless transactions through relayer service •
- Hybrid on-chain/off-chain architecture •

# LITERATURE REVIEW

## Key Findings from Existing Research

Traditional loyalty systems are centralized and fragmented.  
Most existing blockchain loyalty systems use Ethereum/EVM models.  
Existing Web3 loyalty solutions often introduce high UX friction.  
Consortium/private blockchain systems improve control but reduce openness.  
Research on TON-based loyalty architectures is extremely limited.

## No existing work combines:

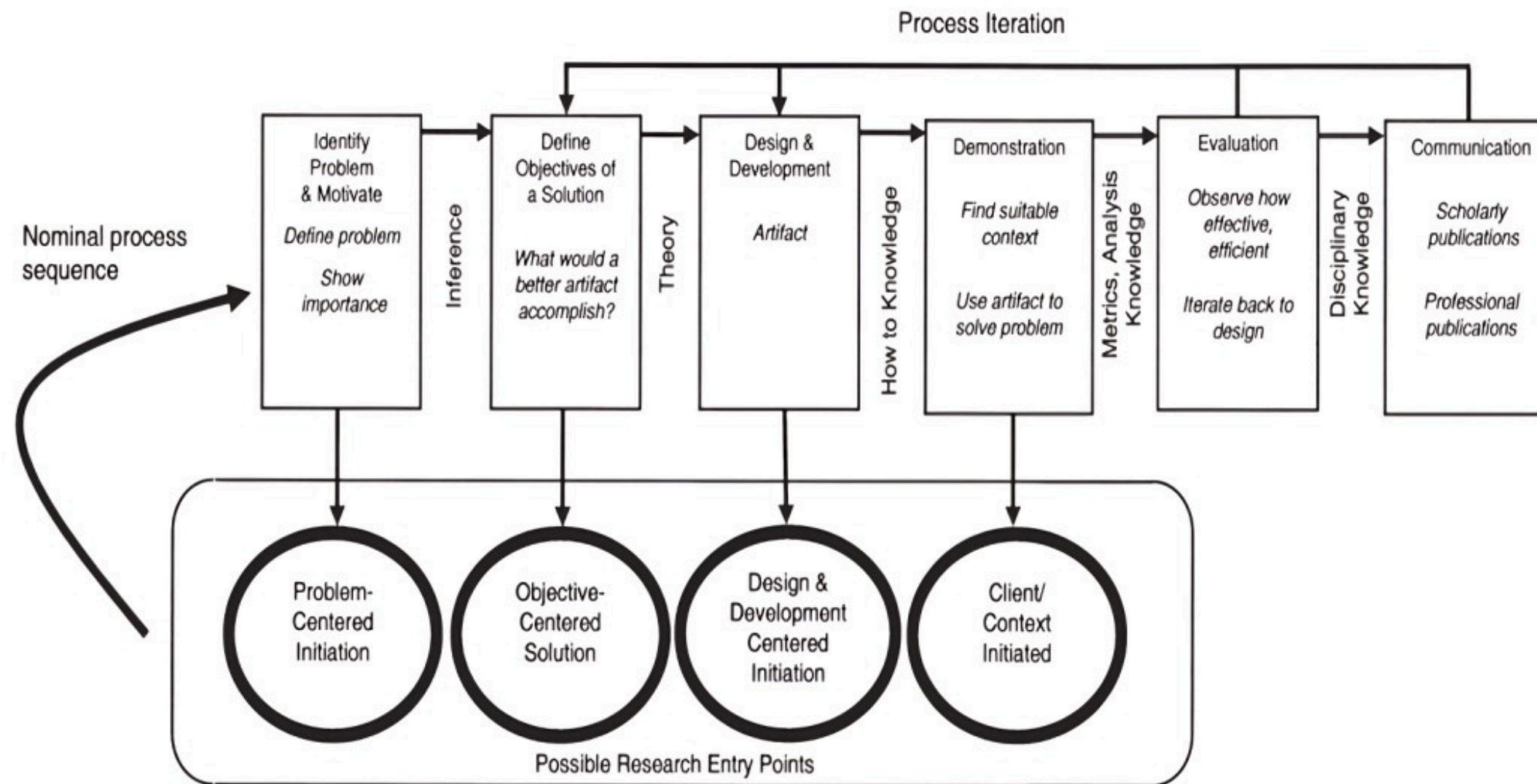
- Public asynchronous blockchains (TON),
- hybrid on-chain/off-chain architecture,
- low-friction B2B loyalty workflows.

| Solution Type                  | Main Limitation       |
|--------------------------------|-----------------------|
| Ethereum loyalty systems       | gas fees, scalability |
| Hyperledger consortium systems | governance overhead   |
| NFT/Web3 loyalty apps          | weak everyday utility |
| Traditional loyalty platforms  | centralized databases |

## Research Gap

**Lack of implementation-oriented architectures for scalable and user-friendly blockchain loyalty ecosystems.**

# METHODOLOGY



- **DSR Framework:** A cyclic process moving from problem identification to technical artifact design and expert evaluation
- **SOR Model:** Modeling how tokenized stimuli (visibility and value) trigger internal states of perceived control to produce loyalty responses

**Figure 2:** The Design Science Research (DSR) cycle used for the iterative development and evaluation of the TON loyalty artifact (adopted from Behrouzi, 2023)



# METHODOLOGY

|                                                |                                                                                            |                                                                                      |
|------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Customer Perception<br/>Pre/Post Survey</b> | Loyalty transparency, Telegram access, trust, and cross-partner utility                    | Mean Likert score comparison for estimation of customer trust in the system          |
| <b>Business Perception<br/>Pre/Post Survey</b> | Partner interest, campaign control, dispute reduction,and reconciliation workload          | Percentage distribution, Likert score change for business friction change estimation |
| <b>Business<br/>Feasibility/KPI</b>            | Transaction volume, basket size, administration cost, platform fee,and payback assumptions | Midpoint conversion of survey categories; net monthly benefit and payback formulas   |
| <b>Architectural<br/>Acceptance</b>            | Security, transparency, availability, scalability, and performance                         | Scenario-based assessment, requirement analysis to evaluate pilot readiness          |

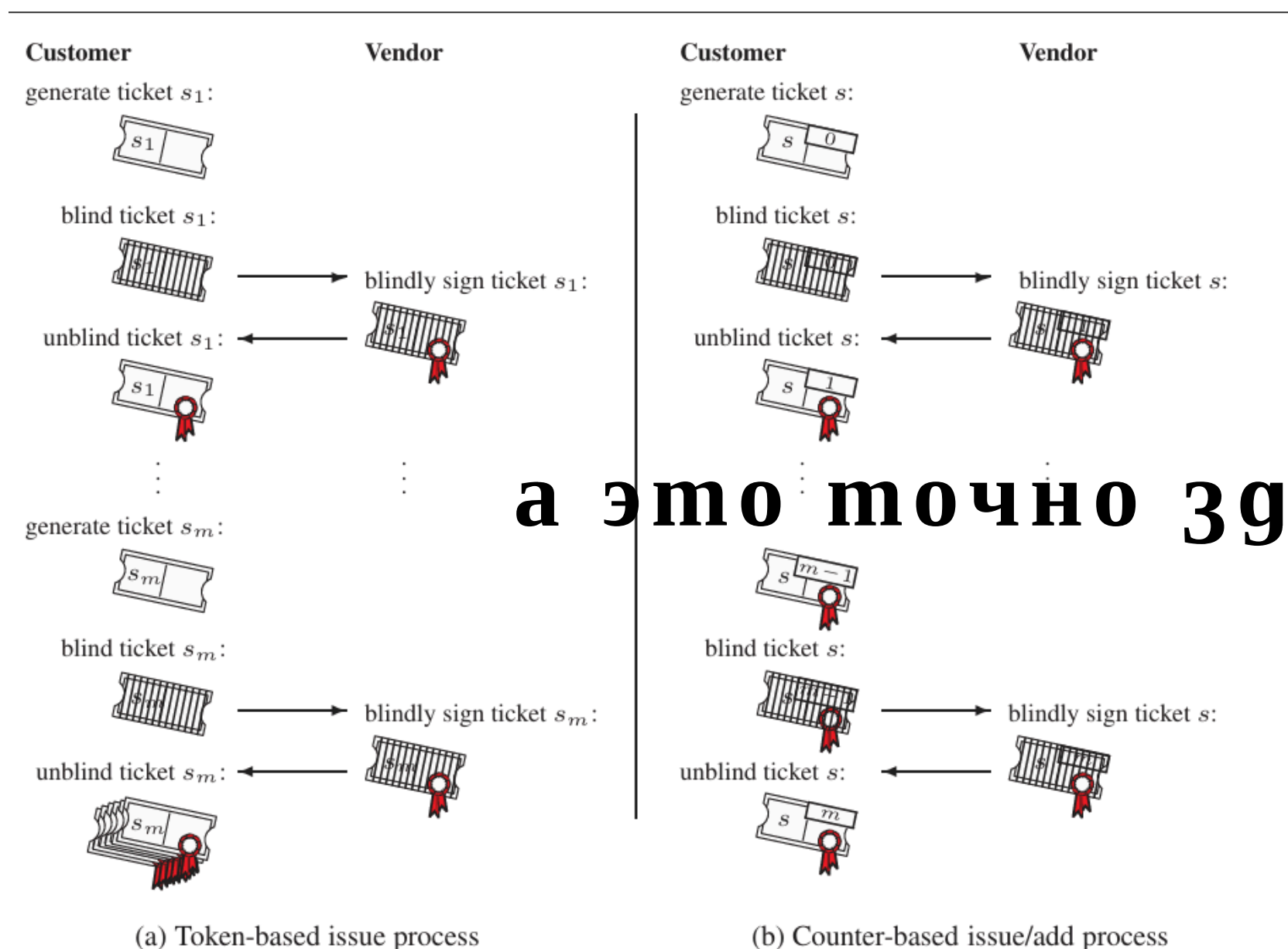
**Table 1:**The criteria/metrics and methods used for evaluation of the platform

This approach ensures reliability of data, as all transformations are explicit and reproducible.Which ensures that it will produce the same results in the same conditions.

The approach covers the main points of friction: customers’ view the loyalty system, businesses’ assesment of system performance, KPI, and architectural compliance in accordance with requirements.

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# Existing Solutions



**Figure 3:** Token-based loyalty system with minimal user-data (adopted from Matthias Enzmann and Markus Schneider Fraunhofer Gesellschaft, 2004)

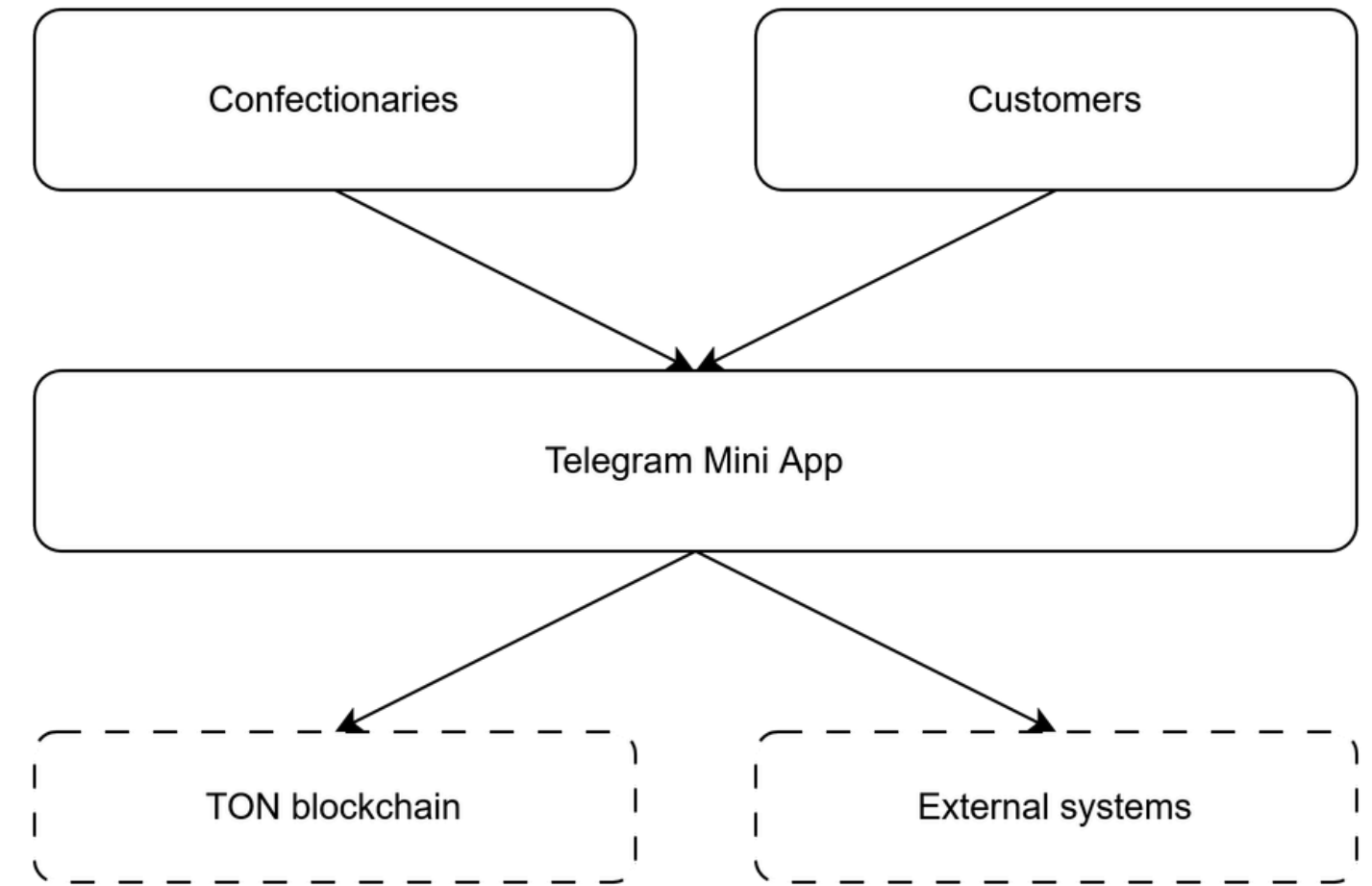
Figure 2 portrays a privacy oriented digital loyalty point issuance system that prevents user information from being tracked during transactions. By applying blinding techniques to user-generated tokens in order to prevent linkage of issuance and redemption of bonus points. This approach has extremely high security and privacy, however it may lack transparency and be overly complex in implementation.

Blockchain-based solutions exist as proof-of-concept projects that lack adoption in the industry.

# PROPOSED SOLUTION

**We propose a B2B loyalty ecosystem on the TON blockchain to resolve market fragmentation and enhance customer psychological ownership by leveraging transparency of blockchain operations.**

- Transition loyalty points from corporate liabilities into liquid digital assets that customers truly own through tokenization (Wang et al., 2019; Santos et al., 2023).
- Establish a generic, "trust-free" platform where multi-brand rewards coexist in a single user interface to eliminate siloed applications (Santos et al., 2023).



**Figure 4:** A high level abstraction of a TON-based confectionery loyalty platform



# PROPOSED SOLUTION

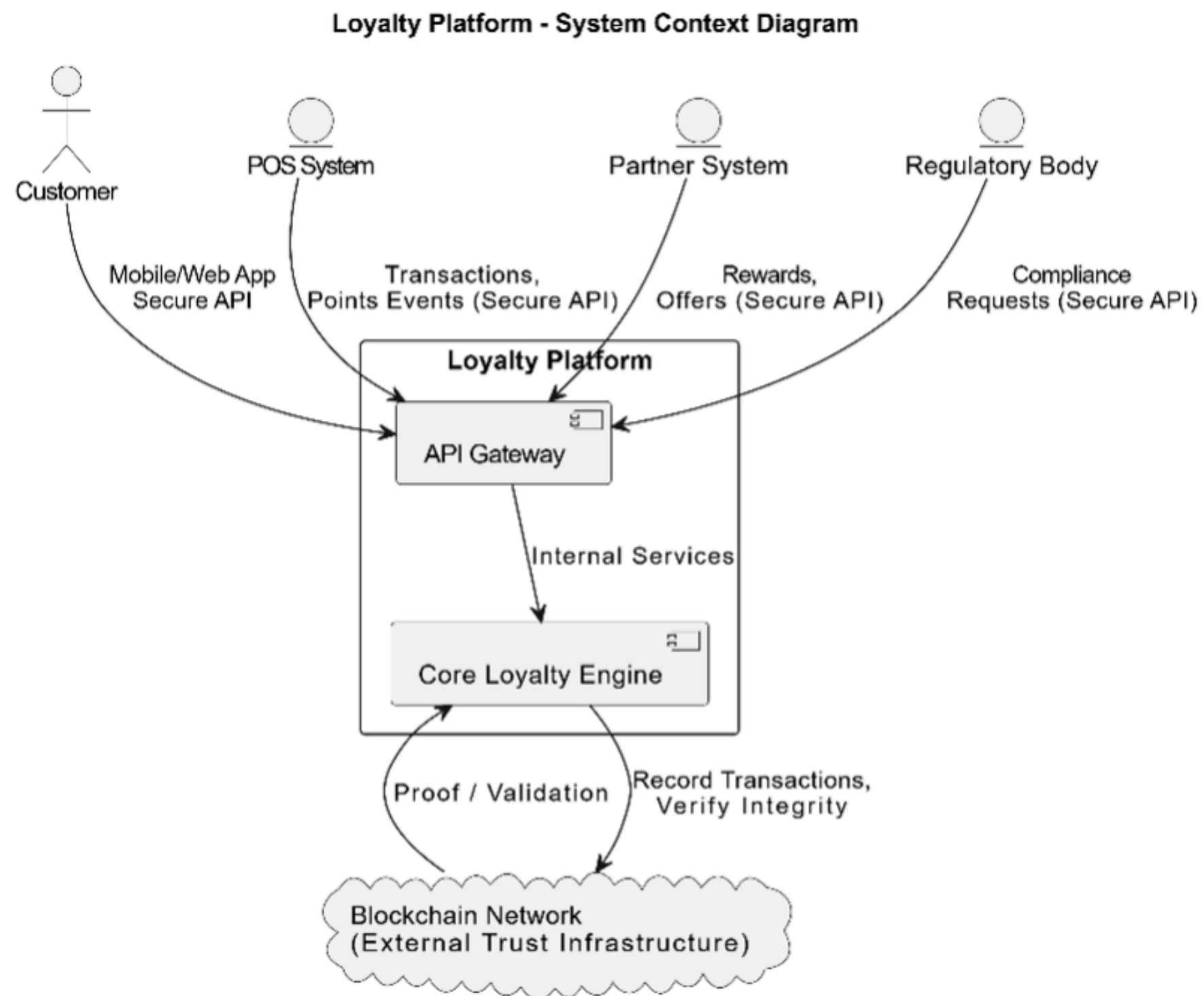
## Sweet Loyalty Platform



### Key Features:

- Blockchain reward tracking
- Automated smart contracts
- Telegram-based access
- Cross-partner ecosystem
- Transparent transactions
- A validation model

# ARCHITECTURE

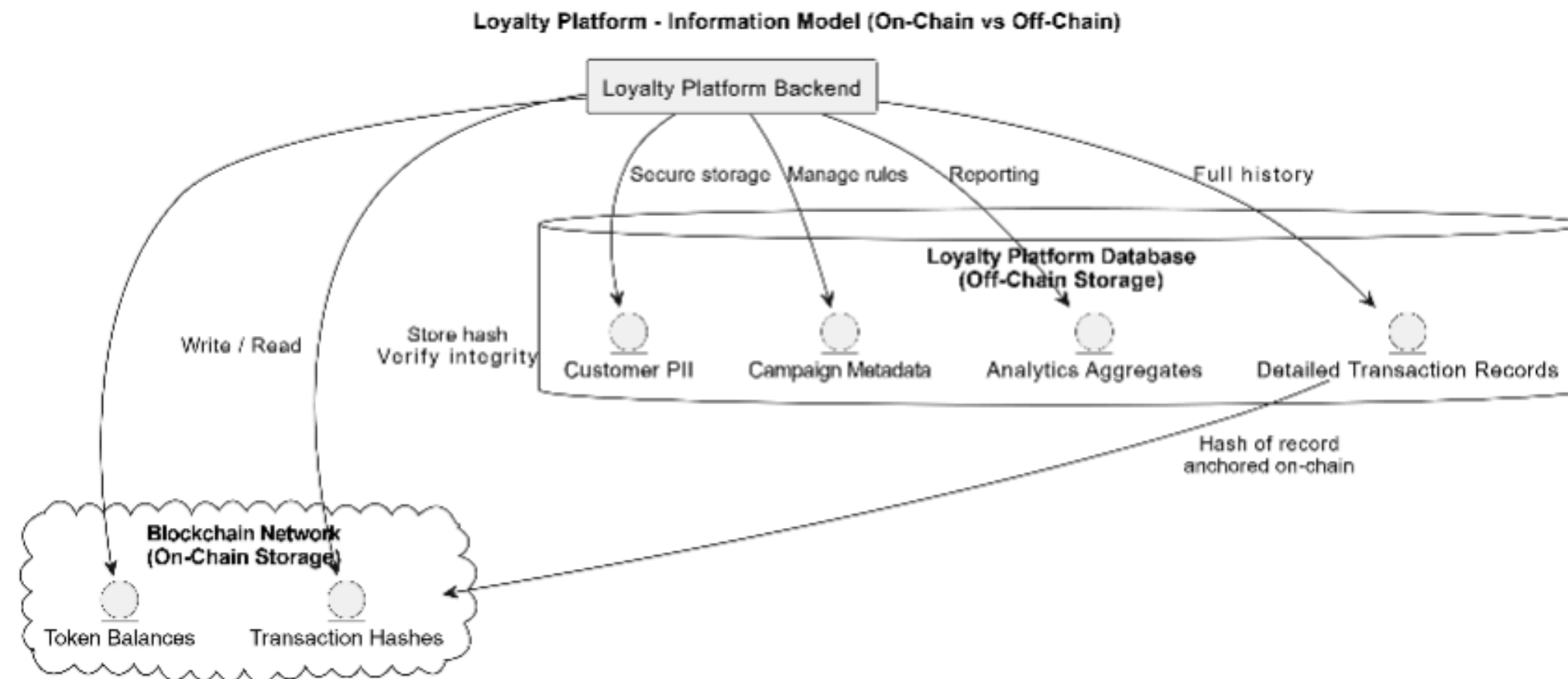


- This view explains component hierarchy and dependencies between components.
- This context diagram represents the actors in the system: Customer, POS system, Partner system.
- The split between internal and external components is clearly defined, and interaction model is presented.

**Figure 4:** Confectionery system context diagram for the proposed loyalty platform.

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# ARCHITECTURE



**Figure 5:** Off- and on-chain informational model of the proposed loyalty platform.

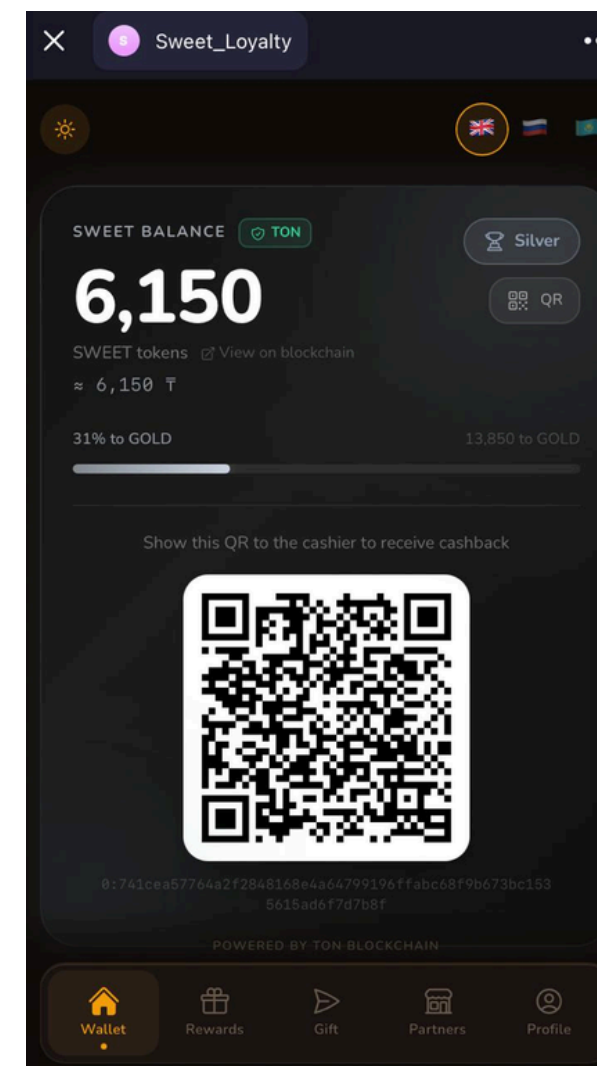
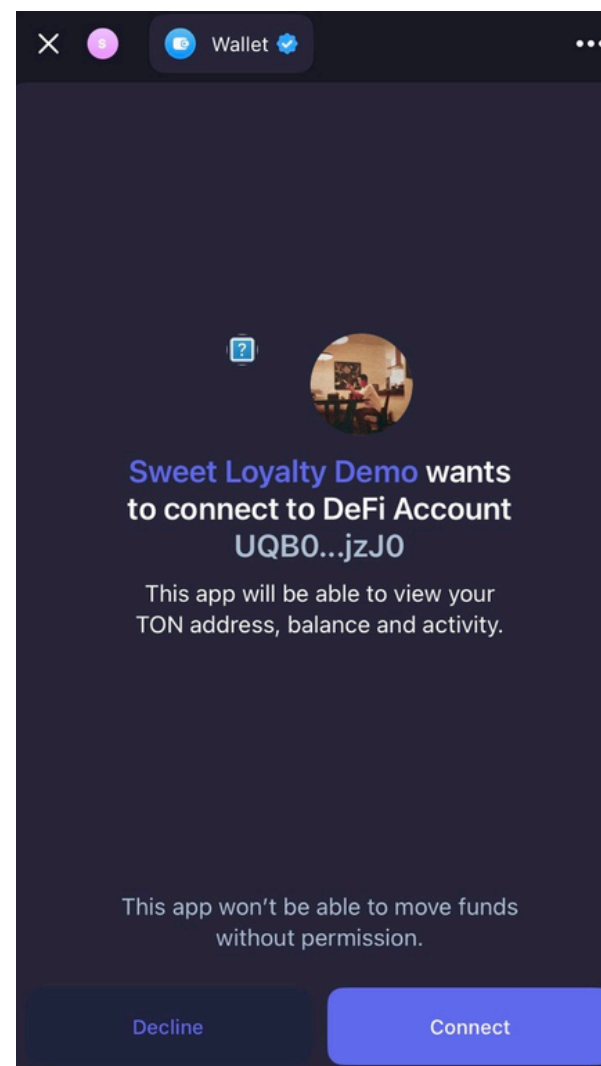
- Figure 5 demonstrates data storage architecture. Data is stored in two ways - on-chain on the TON blockchain, and off-chain in the loyalty platform's databases.
- On-chain storage is used to preserve integrity and transparency of transactions and user balances. Off-chain storage is used to securely store Personally Identifiable Information (PII), analytics, read-only transaction records.

# MVP Demonstration Flow

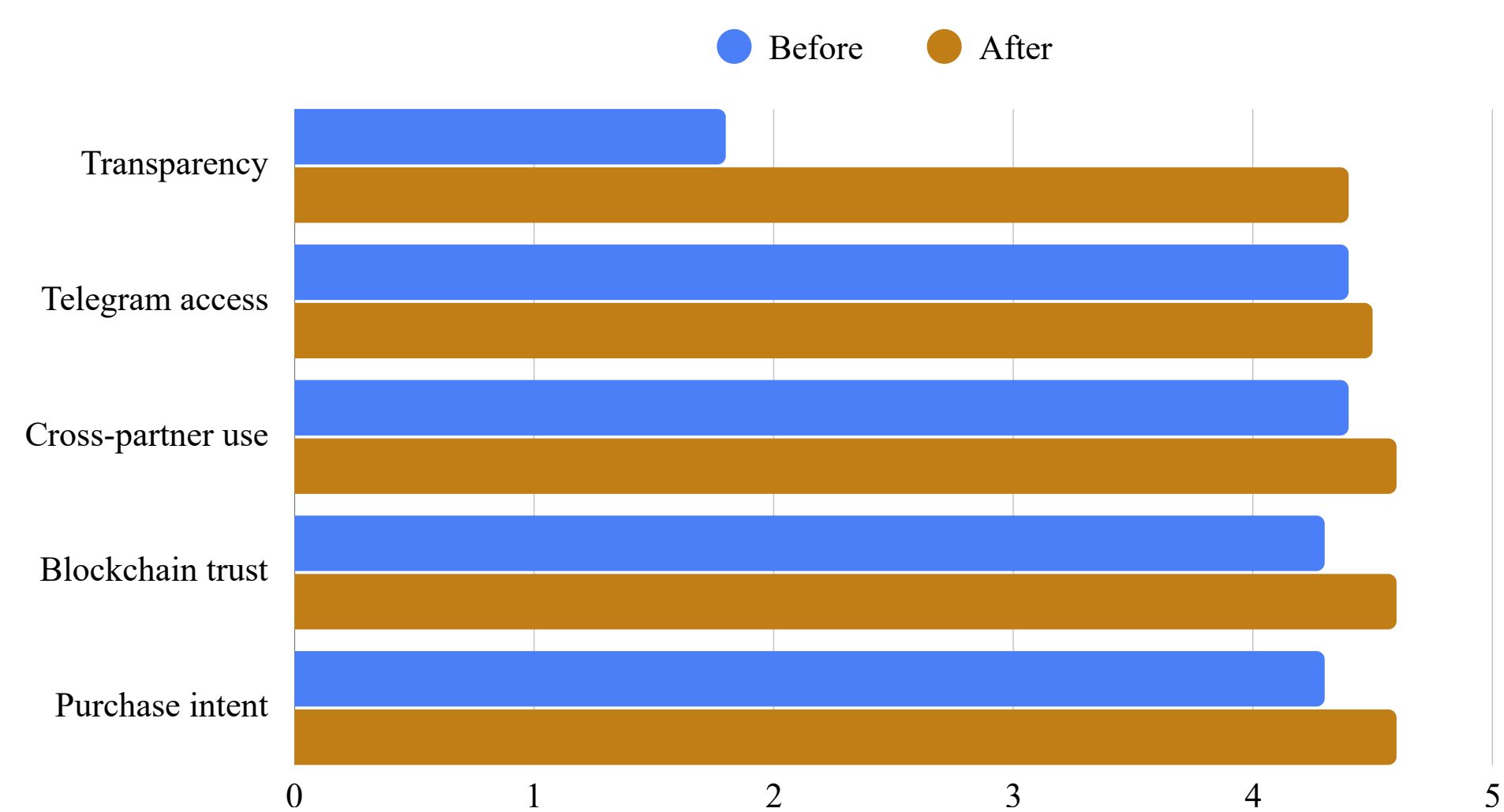
- MVP Scenario

**это лучше в конце пихнуть?**

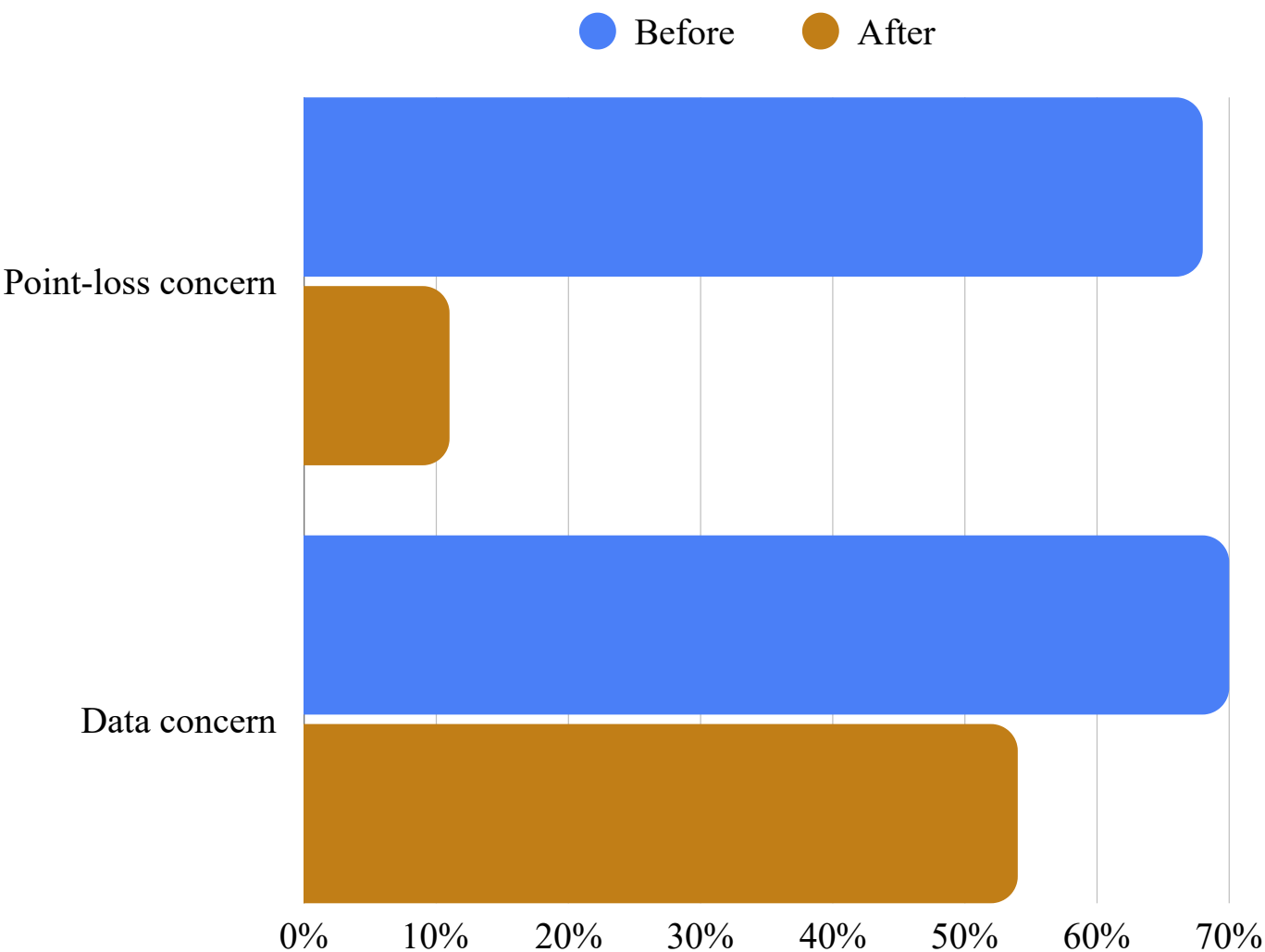
- 1. Customer makes a purchase at a confectionery.
- 2. POS or partner system sends purchase data to the backend.
- 3. Backend validates the transaction and calculates reward points.
- 4. Smart contract records the reward operation on TON.
- 5. Transaction hash is stored for verification.
- 6. Customer sees updated balance in Telegram Mini App.
- 7. Partner can monitor campaign performance and reconciliation data.



# RESULTS (CUSTOMER SIDE)



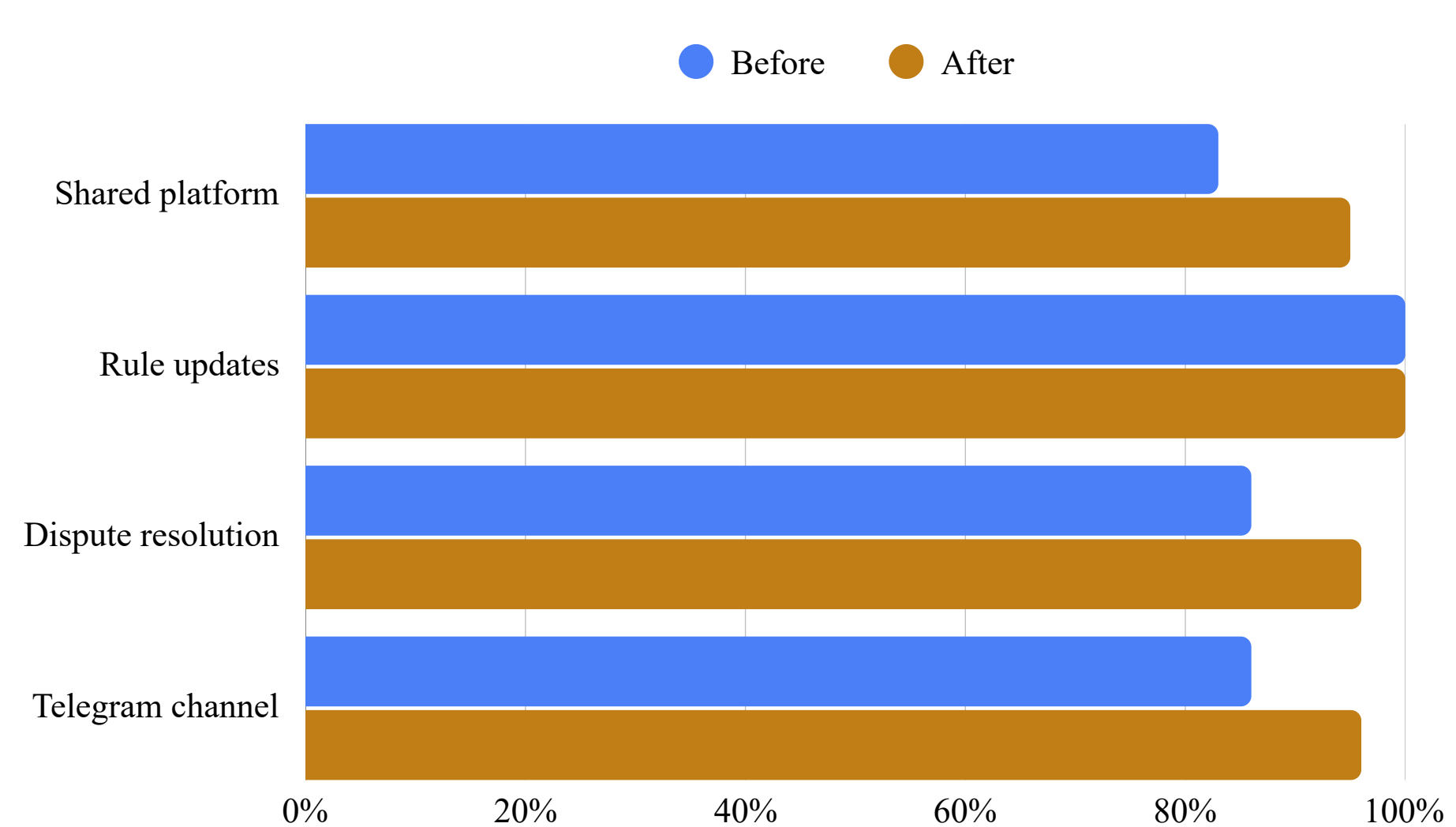
**Figure 6:** Customer perception comparison before and after.



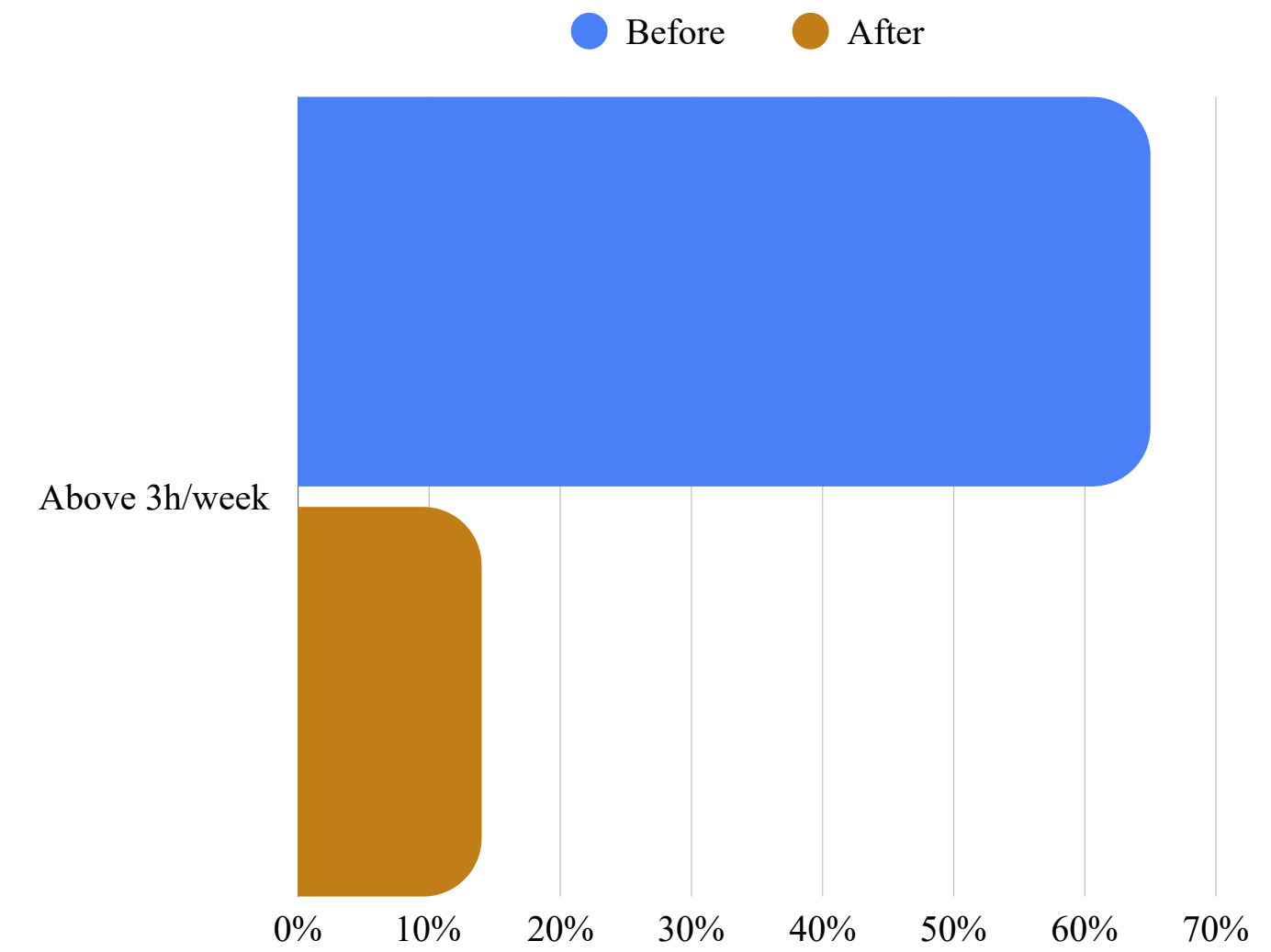
**Figure 7.** Customer risk reduction after MVP demonstration.



# RESULTS (BUSINESS SIDE)

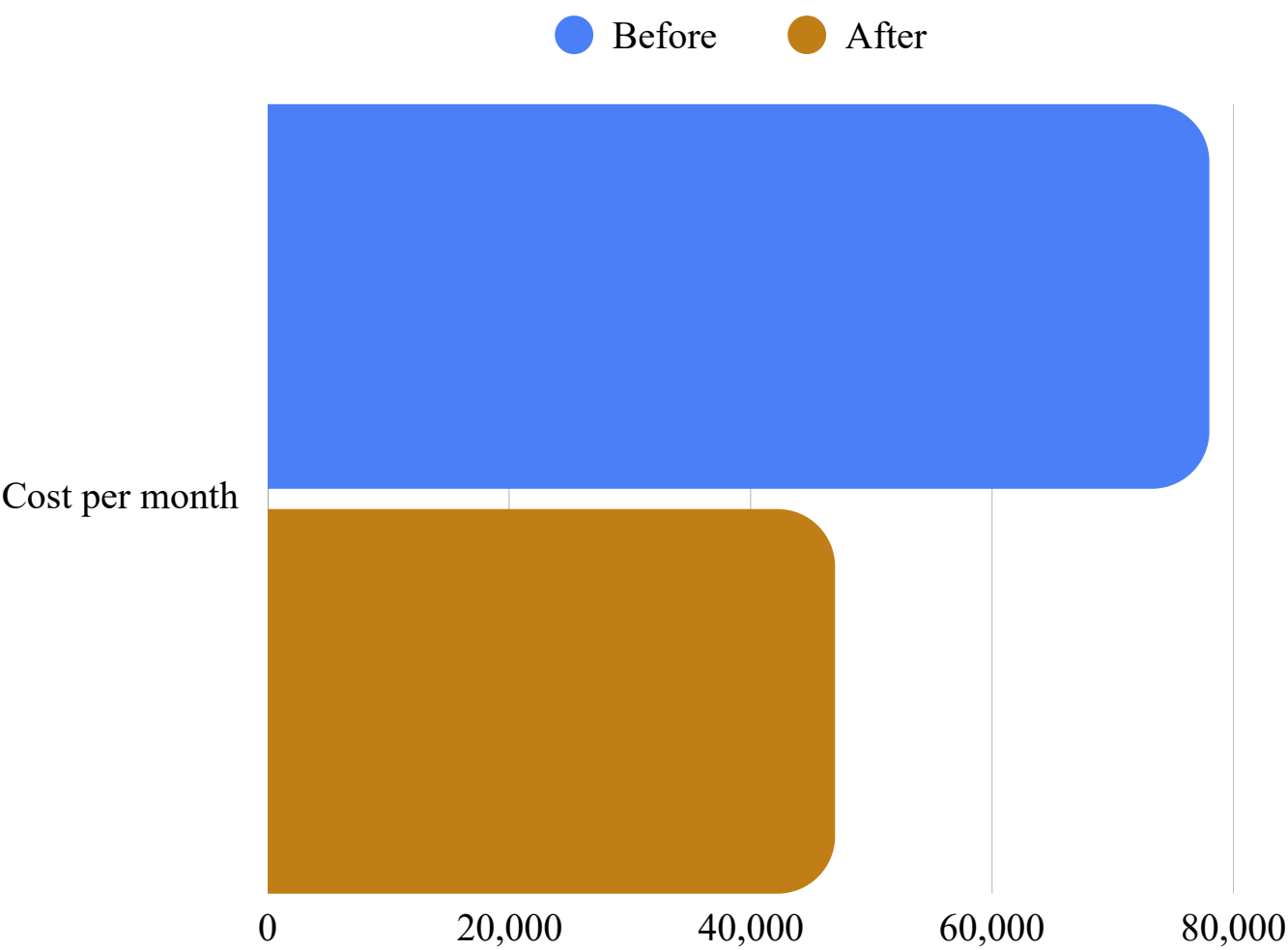


**Figure 8:** Partner-side adoption indicators after MVP demonstration.



**Figure 9.** Manual reconciliation workload before and after MVP demonstration.

# RESULTS (KPI)



**Figure 10.** Average loyalty administration cost before and after MVP demonstration.

|                                     |                      |
|-------------------------------------|----------------------|
| Average net benefit per business    | 159,871.74 KZT/month |
| Average payback period per business | 7.51 months          |
| Pilot reconciliation hours saved    | 268.46 hours/month   |
| Pilot total net benefit             | 3,677,050 KZT/month  |

**Table 2:** Aggregate business KPI model summary.

# CONCLUSION, LIMITATIONS AND FUTURE WORK